

ORIGINAL PAPER

Using Multi-omics Analysis to Determine Novel Biomarker Candidates in Hypermobile Ehlers-Danlos Syndrome (hEDS)

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Abstract

Hypermobile Ehler-Danlos syndrome (hEDS) is a complex disorder mainly characterized by joint hypermobility, chronic pain and fatigue, remaining difficult to diagnose due to a lack of a reliable molecular biomarker. Previous studies have used single-omics-based approaches, but these studies are limited in that the use of only transcriptomics or proteomics limits the complete picture of hEDS-associated mechanisms and insight into overlapping shared results from both omics layers. A multiomics approach can instead link these different biological layers together, enabling patterns between layers and offering a more complete pinpointing of potential biomarkers. Thus, publicly available fibroblast transcriptomics, miRNA, and serum proteomic data were integrated from hEDS patients and controls. The transcriptomic and miRNA expression were checked for differential expression and enriched biological pathways, and the proteomic layer was examined for differential protein abundance and functional enrichment using DAVID. Then, significant biological processes found across the datasets were compared to spot pathway-level convergences. Finally, STRING network analysis was used to explore how the top candidate proteins overlap and connect. Transcriptomic analysis identified 60 significant genes, and 6 genes between the transcriptomic and miRNA-target lists but no genes common overall. Through pathway and network analysis, this study identified a strong AKR1C3 mechanism cluster involving steroid/hormone metabolism, other overlapping genes: STEAP4, KRT7, CDH8, PTPRB, and PLXNC1 that could be tested, as well as overall involvement of pathways in tissue-dysfunction maintenance through immune and tissue related terms. Overall, this provides a foundation for future investigation hEDS biomarkers through validation of suggested candidates.